



East West University

Lab Report

Semester: Summer-2025

Course Title: Electronic Circuits

Course Code: CSE251

Section: 04

Expt No: 04

Expt Name: Introduction to Transistor

Group No: 06

Group Members:

Student Name	Student Id
Md. Hasib Ali	2023-3-60-186
Arijit Kumar Anoy	2023-3-60-055
Md. Shakib Al Hasan	2024-1-60-200
Fahim Shahriar	2023-2-60-094
Morshedul Islam	2023-2-60-125

Submitted by-

Name: Md. Shakib Al Hasan

Id: 2024-1-60-200

Submitted to-

Dr. Sarwar Jahan

Associate Professor

Department of Computer Science & Engineering

East West University

Date of Performance: July 28, 2025

Date of Submission: July 09, 2025

Title: Introduction to Transistor

Objectives:

1. Identify base, emitter and collector terminals and connections of NPN and PNP transistors.
2. Demonstrate and measure the effects on base current of forward and reverse bias in the emitter-base circuit.

Theory (Summary):

It is convenient to represent the current voltage characteristics of transistor graphically. In a BJT common-emitter (CE) configuration, the emitter serves as the common terminal between input and output. The input is applied at the base terminal and the output is taken from the collector terminal. The typical CE output describes i_c as a function of V_{CE} with i_b as a parameter. In the active mode, i_c of practical BJTs shows some dependence on V_{CE} due to early effect. As a result, i_c characteristics are not horizontal straight lines. This dependency of i_c on V_{CE} is included in the equivalent circuit via an output resistance r_o .

Equipment and Component:

1. Power supply
2. Multimeter
3. Resistor ($95.16K\Omega$ and $1.8K\Omega$)
4. Transistor (NPN and PNP)
5. Voltmeter
6. DC milli-ammeter and DC micro-ammeter

Circuit Diagram:

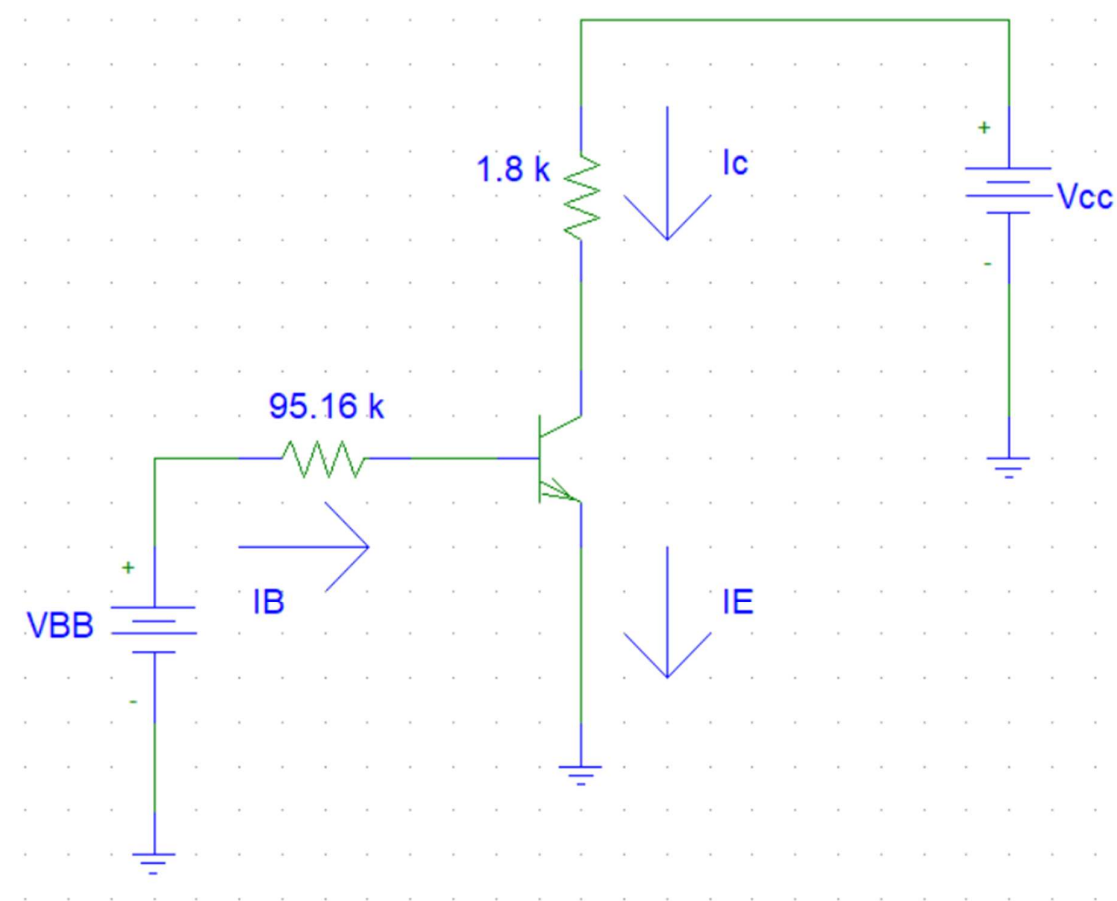


Figure-1: Circuit diagram for measuring I-V characteristics of Transistor

Experimental Datasheet:

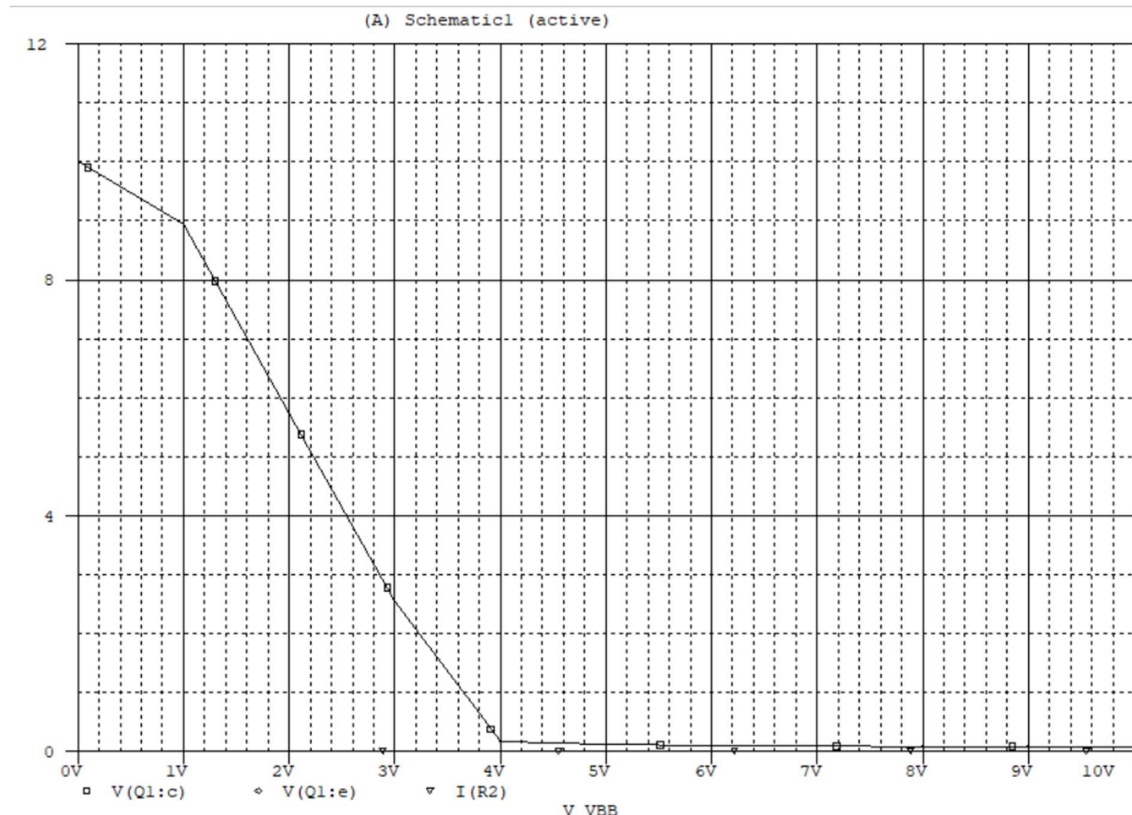
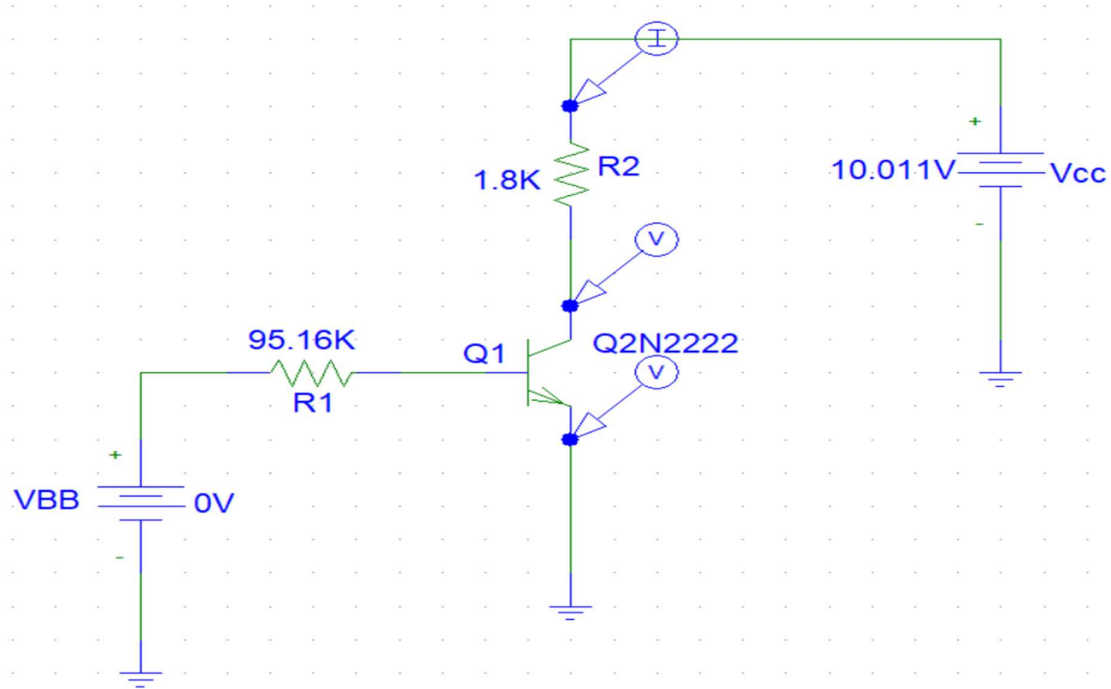
V_{BB}	V_{CC}	V_B	V_C	V_{CE}
10.119V	1	9.510	0.9436	0.0085
	2	9.502	1.8962	0.0146
	3	9.495	2.912	0.201
	4	9.491	3.959	0.0251
	5	9.489	4.975	0.0296
	6	9.487	5.929	0.0335
	7	9.484	6.964	0.0374
	8	9.483	8.007	0.0411
	9	9.481	8.959	0.0442
	10	9.480	9.988	0.0476

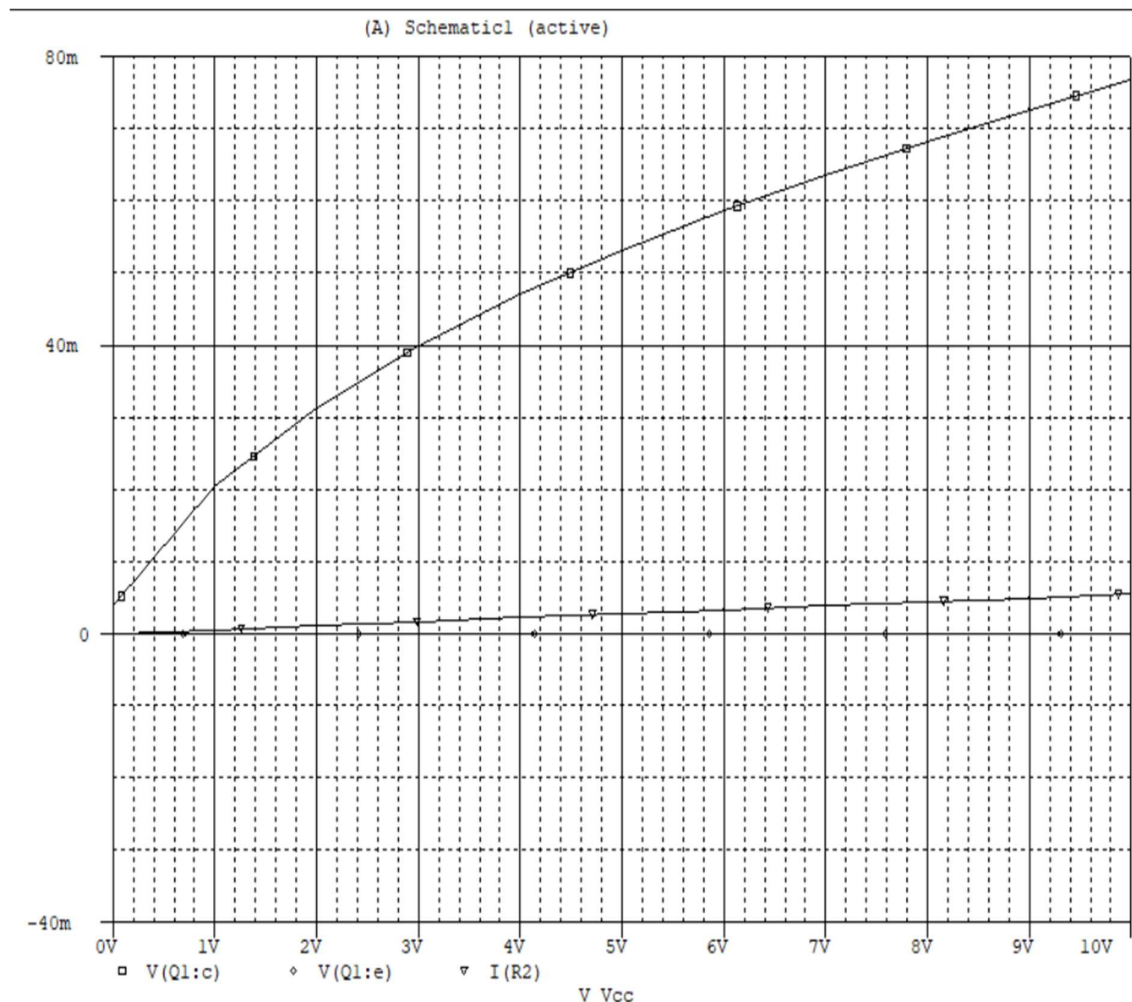
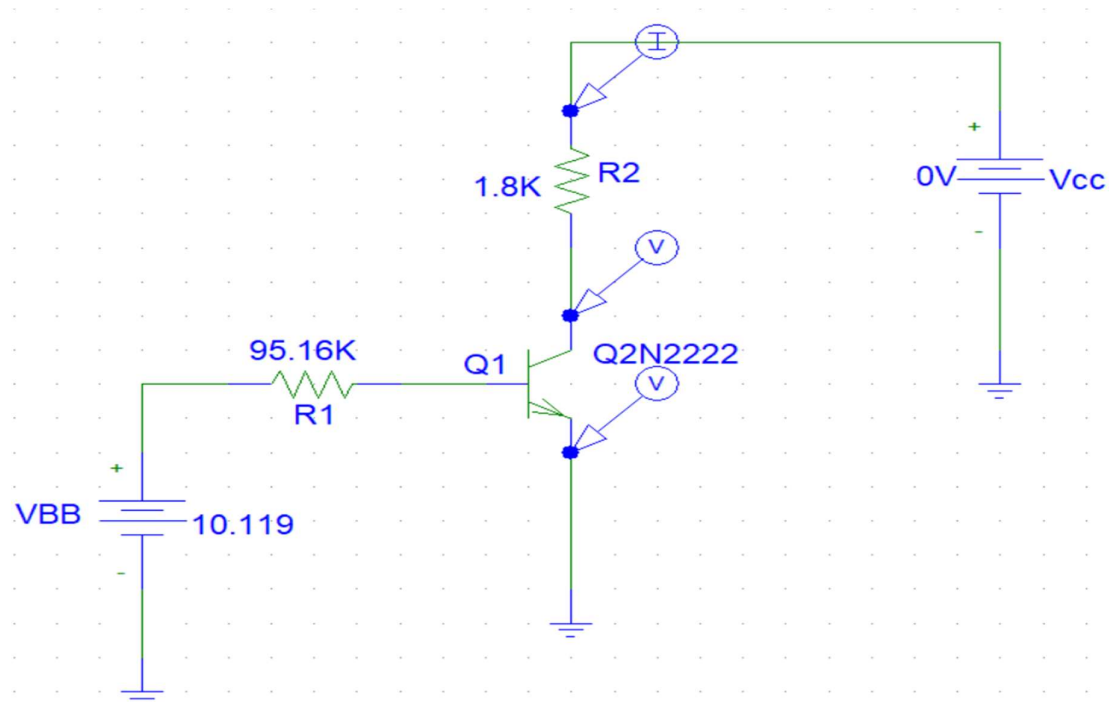
V_{CC}	V_{BB}	V_B	V_C	V_{CE}
10.011V	1	0.3489	1.5382	8.476
	2	1.3058	5.847	4.145
	3	2.985	9.844	0.2058
	4	3.302	9.938	0.0994
	5	4.341	9.973	0.0802
	6	5.317	9.966	0.0698
	7	6.312	9.972	0.0624
	8	7.301	9.982	0.0569
	9	8.412	9.984	0.0519
	10	9.393	10.00	0.0483

Post-Lab:

Plot $I_c - V_{CE}$ characteristics curve from the measured data using PSPICE and MATLAB

Answer





Discussion

In this experiment, we introduced the transistor, a fundamental component in electronic circuits. We explored the basic properties and functioning of a transistor in a common-emitter (CE) configuration, where the emitter is the common terminal for both input and output. The input signal was applied to the base and the output was taken from the collector. The main objective was to understand how a transistor's characteristics are graphically represented by its current voltage (I-V) curves. We also identified the base, emitter and collector terminals and measured the effects of forward and reverse bias on the base current.

The experiment datasheet shows the relationship between V_{BB} , V_{CC} , V_B , V_C and V_{CE} . For a constant V_{CC} of 10.119V, as V_{BB} increases, V_C decreases. Similarly, for a constant V_{CC} of 10.099V, as V_{BB} increases, the V_C also decreases. The graphs, using Pspice, showing the relation between I_C and V_{CE} as the base current changes. Some minor deviations may have occurred due to limitations in measurement accuracy, temperature variations, component tolerances or slight differences in the transistor characteristics. However the overall trends aligned well with theoretical expectations.

Understanding this principals laid for further exploration of electronic circuits, especially in designing amplifiers and circuits, logic gates and switching systems, and signal processing.